

EC2200

Economics of Managerial Decision Making

Winter 2022 Examination · Paper 1

Duration 1.5 hours · Section A: 2 of 3 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2200 — Economics of Managerial Decision Making
Session	Winter 2022
Structure	Section A: three short questions, answer any two (each marked out of its own 100%, contributing 50% of the paper between them). Section B: two long questions, answer one (marked out of 100%, contributing the remaining 50%).
This scheme covers	All three Section A questions and both Section B questions in full.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to the other EC2200 papers. This paper's B1 introduces transfer pricing and the Intel differential-pricing/third-degree price discrimination problem, both distinct to this sitting and among the most substantial numerical questions across the EC2200 papers in this set. A2 extends the consumer theory question seen on the Winter 2024 paper with a new part (c) on the income consumption curve and Engel curve. The decision-making process, Herfindahl Index, price discrimination framework, Porter's Five Forces, and the three stages of production are shared building blocks with consistent underlying content across all the EC2200 papers.

Section A · Question 1

Shareholder Wealth Maximisation, Advertising & Decision-Making · 20+20+20+40%

A1(a)

20%

Outline the main features of the shareholder wealth-maximisation model.

A1(b)

40%

A firm uses two advertising media, TV and Radio. At the current level of advertising, marginal sales revenue generated from a 30 second TV commercial is €150,000. The marginal cost of TV commercial is €50,000. The marginal sales revenue generated from a 30-second radio commercial €12,000. The radio commercial costs €2,000.

(i) What advice would give to the above firm in terms of its advertising strategy?

(ii) Suppose that the marginal cost of a 30 second TV commercial declined to €25,000, how does this impact on the advertising strategy of the firm.

A1(c)

40%

“The ability to make good decisions is the key to successful managerial performance”. Describe and illustrate each step in the decision-making process, clearly labeling each stage in the process. Support your answer with a relevant example.

(a) Main features of the shareholder wealth-maximisation model — 20%

Main features of the shareholder wealth-maximisation model

The objective: maximise the present value (PV) of the firm's expected future profits (or dividends per share), discounted at the shareholders' required rate of return — not simply maximise this period's accounting profit.

$$V_0 = \sum \pi_t / (1+r)^t$$

It is forward-looking and multi-period: the model explicitly values profits expected in every future period, not just the current one, so a decision that sacrifices some profit today for materially higher profit later can still raise firm value.

It incorporates the time value of money: profits expected further in the future are discounted more heavily (at rate r , compounded over t periods) — a euro of profit next year is worth less today than a euro of profit now.

It incorporates risk via the discount rate: the required rate of return (r) reflects the riskiness of the firm's expected profit stream — riskier, more uncertain profit streams are discounted at a higher rate, reducing their contribution to firm value, all else equal.

(b) Advertising strategy: TV vs. Radio — 20% + 20%

Medium	Marginal Revenue (MR)	Marginal Cost (MC)	MR/MC ratio
Television	€150,000	€50,000	3.0
Radio	€12,000	€2,000	6.0

(i) Advice: the optimal allocation of an advertising budget across media equalises the marginal return per euro spent (MR/MC) across all of them. Radio currently returns €6 of marginal revenue per €1 spent, versus only €3 per €1 for TV — radio is the more efficient medium at the margin. **Advice: shift advertising spend from TV toward radio** until the MR/MC ratios are equalised, which raises total profit without necessarily increasing the total advertising budget.

(ii) Impact of TV's marginal cost falling to €25,000: TV's MR/MC ratio becomes $150,000/25,000 = 6.0$, now exactly matching radio's ratio of 6.0. The two media are now equally efficient at the margin, so the current allocation between TV and radio is optimal — **no further reallocation between the two media is advised.**

(c) The decision-making process — 40%

The Managerial Decision-Making Process

Good managerial decisions follow a structured process rather than ad-hoc judgement. The standard steps, illustrated with a worked example (a firm deciding whether to launch a new product line):

- **Step 1 — Define the problem/objective.** State clearly what is being decided and against what criterion it will be judged (usually profit or firm value maximisation). *Example: should the firm launch Product X, judged against its effect on total profit?*
- **Step 2 — Identify the constraints.** The technology, budget, production capacity, regulatory environment, and competitive conditions that limit which choices are actually feasible. *Example: the firm has a fixed production line capacity and a maximum marketing budget for the launch.*
- **Step 3 — Identify the alternatives.** The realistic set of options available given the constraints — e.g. launch now, delay and gather more market research, launch a scaled-down version, or don't launch at all.
- **Step 4 — Evaluate the alternatives at the margin.** For each alternative, compare the marginal (incremental) revenue it would generate against the marginal (incremental) cost it would require — not average or total figures, which can mislead. *Example: compare the additional expected revenue from launching against the additional cost of production and marketing.*
- **Step 5 — Choose the best alternative and implement it.** Select the option that maximises the objective from Step 1, subject to the constraints from Step 2, then put it into action.
- **Step 6 — Monitor outcomes and feed back.** Track the actual results against the decision's expected outcomes, and use this information to inform the next decision (returning to Step 1) — making the process a continuous cycle rather than a one-off exercise.

Why marginal analysis (Step 4) is the crux of the process: a decision should be taken if, and only if, its marginal benefit exceeds its marginal cost — comparing totals or averages can lead to the wrong decision (e.g. a profitable overall product line might still contain an unprofitable marginal unit). This is the same principle underlying the $MR=MC$ profit-maximisation rule used throughout this course.

Mark Allocation — Part (a) (20%)

Tier	Marks	What separates this tier
16–20%	Correctly outlines all four main features: the PV-of-future-profits objective, its forward-looking/multi-period nature, incorporation of the time value of money, and incorporation of risk via the discount rate.	
8–15%	Correctly states the PV objective but misses one or more of the supporting features (timing, risk).	
0–7%	Vague or incorrect description.	

Mark Allocation — Part (b) (20% + 20%)

Tier	Marks	What separates this tier
Full marks	Correctly computes MR/MC ratios for both media, correctly advises reallocating from TV to radio (i), and correctly identifies that the new TV MC equalises the ratios, requiring no further reallocation (ii).	
Partial	Correct direction of advice but reasoning based only on MR–MC rather than the MR/MC ratio.	
Minimal	Incorrect or no meaningful comparison.	

Mark Allocation — Part (c) (40%)

Tier	Marks	What separates this tier
32–40%	All steps of the decision-making process correctly described and illustrated, with a relevant worked example.	
20–31%	Most steps described correctly but the illustration or example is thin.	
0–19%	Partial or generic description.	

Question A1 Total: 100%

Section A · Question 2

Consumer Equilibrium, Price Effects & the Engel Curve · 20+20+20+20+20%

A2(a)

20%

A typical student maximises their utility function $U = U(F, E)$ subject to an income constraint (M). $M = P_F Q_F + P_E Q_E$, where M = money income, F = Food and E = Entertainment, P = price and Q = Quantity. The student has a weekly income of €100, the price of food is €20 per basket and the price of entertainment is €10 per unit.

Illustrate the student's budget line showing consumer equilibrium at 3 baskets of food and 4 units of entertainment.

A2(b)

60%

If the price of food increases from €20 to €25 per basket due to inflation, describe and illustrate typical consumer equilibrium after the price increase.

(i) Identify, using an illustration, the income effect resulting from the increase in the price of food.

(ii) Identify, using an illustration, the substitution effect resulting from the increase in the price of food.

A2(c)

20%

If the student's weekly income increases €100 to €120 and prices remain unchanged, describe and illustrate the student's income consumption curve and derive the Engel curve.

FULL MODEL ANSWER

(a) The budget line and consumer equilibrium — 20%

The budget line: $20F + 10E = 100$. Check: $20(3) + 10(4) = 60 + 40 = \text{€}100$ — confirmed on the budget line.

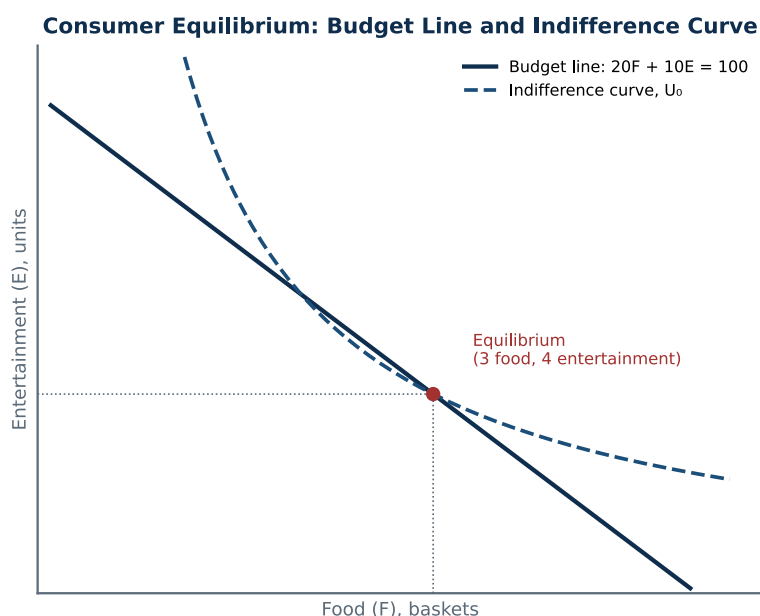


Fig. 1 — The budget line and consumer equilibrium at the tangency with the highest attainable indifference curve

Consumer equilibrium: occurs where the budget line is tangent to the highest attainable indifference curve, i.e. where the price ratio ($-P_F/P_E$) equals the marginal rate of substitution.

(b) The price rise and its decomposition — 20% + 20% + 20%

A rise in the price of food to €25 pivots the budget line inward around its unchanged vertical intercept (10 units of entertainment, since P_E and M are unaffected) — the food-axis intercept falls from 5 baskets to 4. The new equilibrium lies at the tangency with a lower attainable indifference curve.

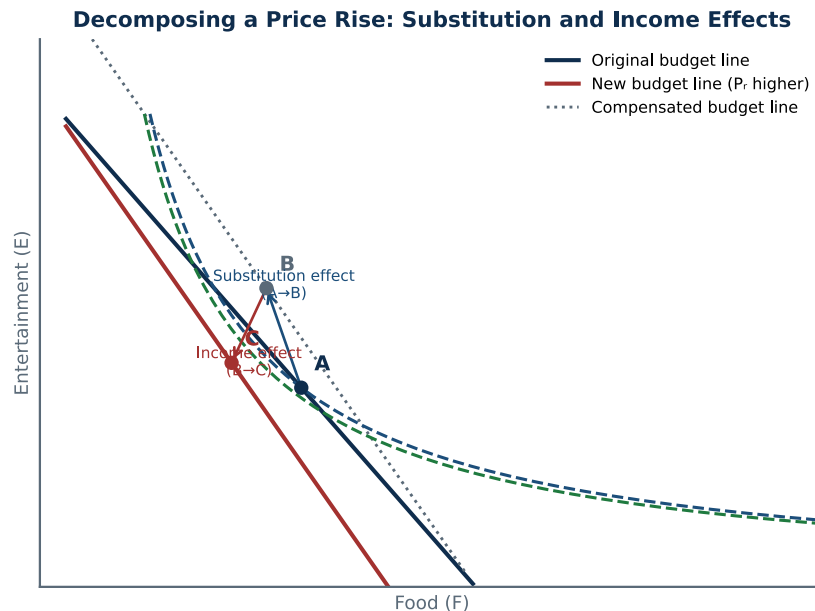


Fig. 2 — Hicksian decomposition: A (original) $\rightarrow$ B (compensated) $\rightarrow$ C (final)

(i) The income effect ($B \rightarrow C$): isolates the fall in real purchasing power caused by the price rise, holding the new (post-rise) relative prices constant — illustrated by shifting the compensated budget line (same slope as the new budget line) until it reaches the actual final budget line. Since food is a normal good, this further reduces food consumption.

(ii) The substitution effect ($A \rightarrow B$): isolates the effect of the change in relative price alone, holding utility (real purchasing power) constant — illustrated by the compensated budget line, drawn with the new, steeper slope but kept tangent to the original indifference curve. Food becomes relatively more expensive, so the student substitutes toward entertainment even before any income effect is considered.

(c) The income consumption curve and the Engel curve — 20%

With prices unchanged ($P_F = €20$, $P_E = €10$) and income rising from €100 to €120, the budget line shifts outward, parallel to itself (same slope, since relative prices haven't changed).

Income Consumption Curve and the Engel Curve

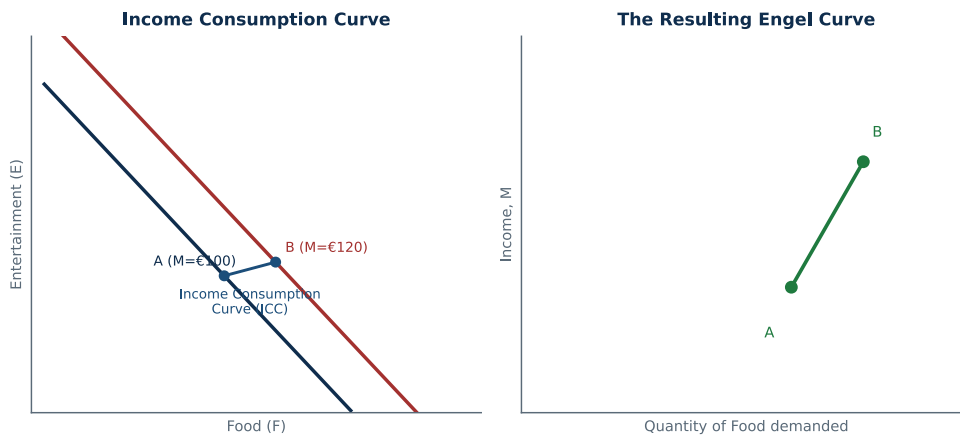


Fig. 3 — The Income Consumption Curve traces equilibria as income rises; the Engel Curve plots income against the resulting quantity of food demanded

The Income Consumption Curve (ICC): traces out the locus of consumer equilibria as income rises with prices held constant — connecting the tangency points on successively higher (parallel-shifted) budget lines as they touch successively higher indifference curves.

Deriving the Engel curve: the Engel curve is derived directly from the ICC by re-plotting the same information in (income, quantity of food) space — for each point on the ICC, read off the income level and the corresponding quantity of food demanded, and plot that pair. Since food is a normal good here (quantity demanded rises from 3 to roughly 3.8 baskets as income rises from €100 to €120), the resulting Engel curve is **upward sloping**. (A Giffen or inferior good would instead produce a downward-sloping, or partly downward-sloping, Engel curve over some income range.)

Exam tip: make sure the Engel curve's axes are correctly labelled — income on one axis, quantity of the specific good (food) on the other — and that its slope is explicitly linked back to whether the good is normal (Engel curve slopes upward) or inferior (slopes downward) over the relevant income range.

Mark Allocation — Part (a) (20%)

Tier	Marks	What separates this tier
16-20%	Correct budget line with correct intercepts, verifies the €100 spend, and correctly shows tangency at the equilibrium.	
8-15%	Correct budget line but tangency/equilibrium condition not clearly shown.	
0-7%	Incorrect or missing budget line.	

Mark Allocation — Part (b) (20% + 20% + 20%)

Tier	Marks	What separates this tier
Full marks each part	Correctly shows the budget line pivot/new equilibrium, and correctly constructs the Hicksian decomposition for both the income and substitution effects with clear labelling.	
Partial	Correct general direction but the decomposition mechanics (compensated budget line construction) are unclear.	
Minimal	Income and substitution effects not meaningfully distinguished.	

Mark Allocation — Part (c) (20%)

Tier	Marks	What separates this tier
16-20%	Correctly illustrates the ICC as a locus of equilibria along parallel-shifted budget lines, and correctly derives an upward-sloping Engel curve with correctly labelled axes, explicitly linked to food being a normal good.	
8-15%	Correct ICC or Engel curve shown, but the derivation link between the two, or the normal-good interpretation, is unclear.	
0-7%	Minimal or incorrect treatment.	

Question A2 Total: 100%

Section A · Question 3

Market Concentration & Porter's Five Forces · 20+30+50%

A3(a)

50%

Market share distributions for US Automobile Industry for the top 6 firms are presented for 2020 in percentage terms in Table A1: General Motors 22%, Toyota 20%, Ford 17%, Honda 15%, Nissan 10%, Others 16%.

(i) Calculate the Herfindahl (H) Index for the US Automobile Industry based on the data in Table A1.

(ii) Interpret the H-Index as an indicator of the amount of competition amongst the firms.

A3(b)

50%

Describe, with the aid of an illustration, the main features of Porter's Five Forces Strategic Model and discuss how the threat of new entrants is dealt with by Porter.

FULL MODEL ANSWER

(a) The Herfindahl Index — 20% + 30%

Firm	Market share (%)	Share ²
General Motors	22	484
Toyota	20	400
Ford	17	289
Honda	15	225
Nissan	10	100
Others	16	256
HHI (sum of squared shares)		1,754

(ii) Interpretation: at 1,754, this falls in the **moderately concentrated** range (1,500–2,500) under the standard antitrust thresholds — a market with meaningful competition among several sizeable firms (the largest, GM, holds only 22%), but concentrated enough that the top five named firms account for the large majority of total sales.

Caveat: the 'Others' category (16%) likely represents several smaller firms grouped together rather than one entity, which overstates their true combined contribution to the HHI — the actual HHI, with 'Others' broken into its constituent firms, would likely be somewhat lower than 1,754.

(b) Porter's Five Forces — 50%

Porter's Five Forces Strategic Model

Porter's Five Forces Strategic Model

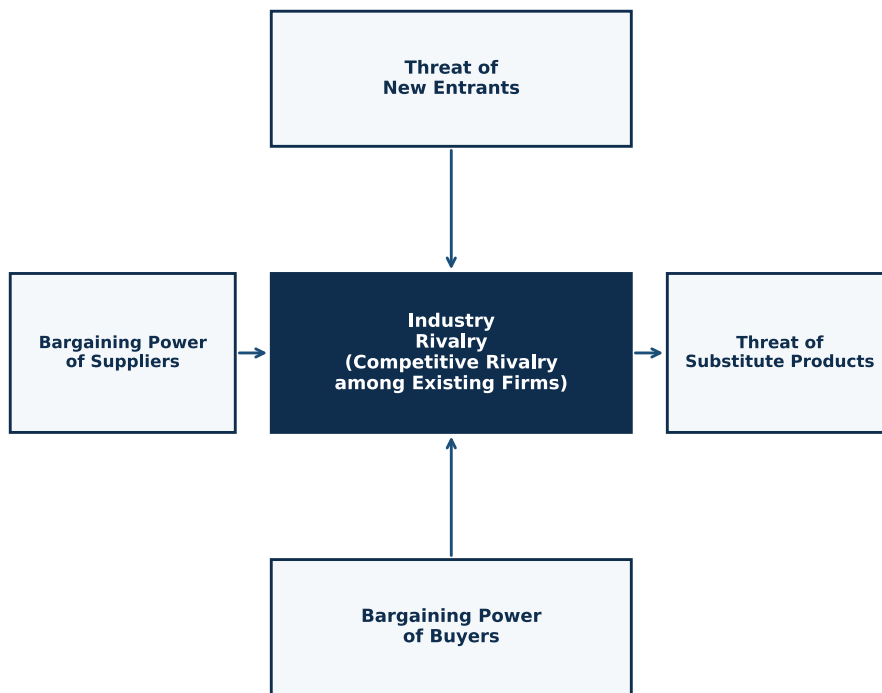


Fig. — The five forces shaping industry competitiveness and profitability

Michael Porter's framework identifies five competitive forces that jointly determine the intensity of competition — and hence the long-run profitability — of an industry. The stronger each force, the more it erodes the potential for firms in the industry to earn above-normal profit.

- **Threat of new entrants:** the ease with which new firms can enter the industry and compete away incumbents' profits — governed largely by barriers to entry (discussed in detail below).
- **Bargaining power of buyers:** the ability of customers to negotiate lower prices or better terms — strong when buyers are large/concentrated, purchases are a large share of buyer spending, products are undifferentiated, or switching suppliers is easy.
- **Bargaining power of suppliers:** the ability of input suppliers to raise prices or reduce quality/terms — strong when suppliers are concentrated, inputs are differentiated or have no substitutes, or switching suppliers is costly.
- **Threat of substitute products:** the availability of alternative products or services that satisfy the same underlying customer need, which caps how much firms in the industry can charge before customers switch away entirely.
- **Industry rivalry (competitive rivalry among existing firms):** the intensity of head-to-head competition among the firms already in the industry — higher when there are many similarly-sized competitors, industry growth is slow, products are undifferentiated, or exit barriers are high.

How Porter addresses the threat of new entrants specifically

Porter argues that the threat of new entrants depends on the height of **barriers to entry** — structural features of the industry that make it costly or difficult for a new firm to enter and compete effectively, even if the industry is currently profitable:

- **Economies of scale:** if incumbents produce at a large enough scale that average costs are much lower than a new, small entrant could achieve, the entrant is forced to choose between entering at a large (risky, capital-intensive) scale or accepting a cost disadvantage.
- **Capital requirements:** industries requiring very large upfront investment (factories, R&D, brand-building) deter entry by all but the best-resourced potential competitors.
- **Product differentiation and brand loyalty:** where incumbents have

established strong brand identity and customer loyalty, a new entrant must spend heavily on marketing simply to become a credible alternative.

- **Switching costs:** if customers face a real cost (financial, time, or effort) in switching from an incumbent to a new entrant's product, the entrant must overcome this inertia, typically by offering large price or quality advantages.
- **Access to distribution channels:** if existing distribution channels (retail shelf space, wholesalers, platforms) are already tied up by incumbents, a new entrant may need to create its own channels — a significant additional cost and barrier.
- **Government policy and regulation:** licensing requirements, patents, and other regulatory barriers can legally restrict entry regardless of a potential entrant's efficiency.
- **Expected retaliation:** if incumbents have a credible history or capacity for aggressive retaliation against new entrants (price wars, capacity expansion), this deters entry even absent any structural barrier, since a rational entrant anticipates the cost of a fight.

Porter's overall point: the threat of new entrants is not simply about whether entry is legally possible, but about how high these structural and strategic barriers are — an industry can be highly profitable yet remain uncontested if barriers to entry are high enough that a rational new entrant expects entry to be unprofitable even against currently high incumbent margins.

Mark Allocation — Part (a) (20% + 30%)

Tier	Marks	What separates this tier
Full marks	Correct HHI calculation (1,754) with all six terms shown, and correct interpretation using the standard concentration thresholds.	
Partial	Correct calculation but limited interpretation, or a minor arithmetic slip.	
Minimal	Incorrect HHI formula or no meaningful interpretation.	

Mark Allocation — Part (b) (50%)

Tier	Marks	What separates this tier
38-50%	Correctly describes all five forces with a correct diagram, and gives a thorough discussion of at least four distinct barriers to entry.	
25-37%	Describes the five forces reasonably well but the threat-of-new-entrants discussion is limited.	
10-24%	Partial or generic description.	
0-9%	Minimal or largely incorrect treatment.	

Question A3 Total: 100%

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full

Question B1 below is the most substantial numerical question across the EC2200 papers in this set — a full third-degree price discrimination problem for Intel selling into two markets, followed by the effect of a legal ban on that price discrimination.

Section B · Question 1(a)

Transfer Pricing · 20%

B1(a)**20%**

Describe the main features of transfer pricing. Support your answer with a relevant example of a firm operating transfer pricing.

FULL MODEL ANSWER**Main features of transfer pricing**

What it is: transfer pricing is the pricing of goods, services, or intellectual property transferred between different divisions or subsidiaries of the same firm (commonly across international borders within a multinational group), rather than sold to an external, independent customer.

- **Purpose — internal resource allocation and performance measurement:** transfer prices let a multinational evaluate the profitability of each individual division separately, even though the divisions are all part of one overall firm, by attributing revenue to the selling division and cost to the buying division.
- **Basis for setting the price:** transfer prices can be set at market price (if an external market for the good exists), at cost (or cost-plus-markup), or negotiated between divisional managers — the choice affects each division's reported profit and the incentives divisional managers face.
- **Tax and profit-shifting implications:** because transfer prices determine how much profit is booked in each division/jurisdiction, multinationals have an incentive to set transfer prices so as to shift more profit into lower-tax jurisdictions — a major reason transfer pricing is closely regulated.
- **The 'arm's length principle':** tax authorities (following OECD guidelines) generally require that transfer prices approximate what unrelated (independent) parties would have agreed for a similar transaction, precisely to prevent the profit-shifting/tax-avoidance abuse described above.

Example: a multinational technology firm manufactures semiconductor chips at a production subsidiary in Ireland and 'sells' them internally, at an internal transfer price, to its own marketing and distribution subsidiaries in other countries (echoing the Intel manufacturing-in-Ireland scenario examined elsewhere on this paper) — the transfer price used for that internal sale determines how much profit is recorded in Ireland versus in the destination country, which is exactly why such prices are subject to arm's-length regulatory scrutiny.

Mark Allocation (20%)

Tier	Marks	What separates this tier
16-20%	Correctly describes at least three features of transfer pricing (internal performance measurement, pricing basis, tax/profit-shifting implications, the arm's-length principle) with a relevant, correctly-explained example.	
8-15%	Correct basic definition with one or two features described.	
0-7%	Vague or incorrect description.	

Section B · Question 1(b)

Intel's Differential Pricing: Japan vs. US · 20+10+10+10%

B1(b)

Intel build computer chips in Ireland, which it ships to computer manufacturers in Japan (Market 1) and the United States (Market 2). Intel practices differential pricing. Demand for chips in the two markets is given by the following functions:

$$\text{Japan: } P_1 = 12 - Q_1 \quad \text{United States (US): } P_2 = 8 - Q_2$$

where Q_1 and Q_2 are respective quantities sold (in millions of units) and P_1 and P_2 are the respective prices (\$ millions) in the two markets. Intel's total cost (TC) function (\$ millions) for these memory chips is: $TC = 5 + 2(Q_1 + Q_2)$.

B1(b)(i)**20%**

Calculate the profit maximising level of output in both the Japanese and US markets.

B1(b)(ii)**10%**

Calculate the per unit price for memory chips in each market.

B1(b)(iii)**10%**

Calculate Intel's combined profit from the Japan and US markets.

B1(b)(iv)**10%**

Determine the price elasticity of demand in both the Japanese and US markets and interpret the elasticity values in terms of the respective markets.

FULL MODEL ANSWER

This is a textbook **third-degree price discrimination** problem (see the general framework covered elsewhere on this paper/set): Intel can identify and separate the two markets (chips are shipped to different manufacturers in different countries, and resale between them is impractical), so it maximises profit separately in each market by setting $MR = MC$ in each, where MC is the same (constant, at €2m per unit) in both, since production is centralised in Ireland.

(i) Profit-maximising output in each market — 20%

Step	Japan (Market 1)	US (Market 2)
Demand	$P_1 = 12 - Q_1$	$P_2 = 8 - Q_2$
Total revenue	$TR_1 = 12Q_1 - Q_1^2$	$TR_2 = 8Q_2 - Q_2^2$
Marginal revenue	$MR_1 = 12 - 2Q_1$	$MR_2 = 8 - 2Q_2$
Marginal cost (from TC)	$MC = 2$ (both markets)	$MC = 2$ (both markets)
Set $MR = MC$ and solve	$12 - 2Q_1 = 2 \Rightarrow Q_1^* = 5$	$8 - 2Q_2 = 2 \Rightarrow Q_2^* = 3$

(ii) Price in each market — 10%

Market	Working	Price
Japan	$P_1 = 12 - Q_1^* = 12 - 5$	\$7m
US	$P_2 = 8 - Q_2^* = 8 - 3$	\$5m

(iii) Combined profit — 10%

Step	Working
Total revenue	$TR = P_1Q_1 + P_2Q_2 = (7 \times 5) + (5 \times 3) = 35 + 15 = \$50m$
Total cost	$TC = 5 + 2(Q_1 + Q_2) = 5 + 2(5 + 3) = 5 + 16 = \$21m$
Combined profit	$\pi = TR - TC = 50 - 21 = \\$29m$

(iv) Price elasticity of demand in each market — 10%

Market	Working ($E_d = (dQ/dP) \times (P/Q)$)	Elasticity
Japan	$dQ_1/dP_1 = -1 \Rightarrow E_{d1} = -1 \times (7/5)$	-1.40
US	$dQ_2/dP_2 = -1 \Rightarrow E_{d2} = -1 \times (5/3)$	-1.67

Interpretation: demand is elastic in both markets ($|E_d| > 1$), but demand in the **US is more elastic** (-1.67) than in **Japan** (-1.40) — US buyers are relatively more price-sensitive. This is exactly consistent with the standard third-degree price discrimination result: the market with the less elastic demand (Japan) is charged the higher price (\$7m), and the market with more elastic demand (the US) is charged the lower price (\$5m) — a profit-maximising price discriminator always charges more in the market where customers are less sensitive to price.

Mark Allocation — Part (i) (20%)

Tier	Marks	What separates this tier
16-20%	Correct TR/MR derivation for both markets, correct constant MC=2 identified from the TC function, and correctly solved $Q_1^*=5$, $Q_2^*=3$.	
8-15%	Correct method with an arithmetic slip in one market carried through consistently.	
0-7%	Incorrect MR derivation or incorrect MC identification.	

Mark Allocation — Part (ii) (10%)

Tier	Marks	What separates this tier
8-10%	Correct $P_1=\$7m$ and $P_2=\$5m$, correctly substituted from part (i)'s quantities.	
0-7%	Incorrect substitution or own-figure error from part (i) not carried through correctly.	

Mark Allocation — Part (iii) (10%)

Tier	Marks	What separates this tier
8-10%	Correct combined TR, correct TC using the given TC function, and correct profit of \$29m.	
0-7%	Incorrect TR/TC construction.	

Mark Allocation — Part (iv) (10%)

Tier	Marks	What separates this tier
8-10%	Correct elasticity formula applied to both markets with correct values, and a correct interpretation linking lower elasticity to the higher price (Japan) and vice versa.	
4-7%	Correct elasticity values but limited or missing interpretation.	
0-3%	Incorrect elasticity calculation.	

Section B · Question 1(c)

The Effect of Banning Price Discrimination · 10+20%

B1(c)

Intel is prohibited by legislation from engaging in differential pricing and price discrimination is no longer possible. The price in both markets is now the same.

B1(c)(i)**10%**

Calculate the impact of this legislation on the price and output in both the US and Japanese markets.

B1(c)(ii)**20%**

Determine the impact of this legislation on Intel's combined profits.

FULL MODEL ANSWER**(i) The uniform price and combined output — 10%**

With a single price required across both markets, Intel now maximises profit against the combined (aggregate) demand curve, found by horizontally summing the two markets' demand curves over the price range where both are served: for $P < 8$ (below the US market's maximum willingness to pay), both markets buy, so $Q_{\text{total}} = (12 - P) + (8 - P) = 20 - 2P$.

Step	Working
Invert combined demand	$Q_{\text{total}} = 20 - 2P \Rightarrow P = 10 - Q_{\text{total}}/2$
Total revenue	$TR = P \times Q_{\text{total}} = 10Q_{\text{total}} - Q_{\text{total}}^2/2$
Marginal revenue	$MR = 10 - Q_{\text{total}}$
Set $MR = MC = 2$ and solve	$10 - Q_{\text{total}} = 2 \Rightarrow Q_{\text{total}}^* = 8$
Uniform price	$P^* = 10 - 8/2 = \\$6\text{m}$

Checking both markets are actually served at $P = \$6\text{m}$: Japan: $Q_1 = 12 - 6 = 6$ million units (positive, valid). US: $Q_2 = 8 - 6 = 2$ million units (positive, valid) — confirming the assumption that both markets remain served at the uniform price.

	Under price discrimination (Q. B1(b))	Under uniform pricing (banned)	Change
Price	Japan \$7m / US \$5m	\$6m in both markets	Japan: ↓ \$1m US: ↑ \$1m
Output	Japan 5m / US 3m units	Japan 6m / US 2m units	Japan: ↑ 1m units US: ↓ 1m units
Total output	8m units	8m units	Unchanged

Interpretation: banning price discrimination doesn't change total output at all (8m units either way) — it just redistributes who buys how much. Japan (previously charged the higher, less-elastic-demand price) now pays less and buys more; the US (previously charged the lower, more-elastic-demand price) now pays more and buys less. This is the standard result: uniform pricing effectively cross-subsidises the previously higher-priced market at the expense of the previously lower-priced one.

(ii) Impact on Intel's combined profit — 20%

Step	Working
Total revenue (uniform pricing)	$TR = P \times Q_{\text{total}} = 6 \times 8 = \48m
Total cost	$TC = 5 + 2(8) = 5 + 16 = \21m
Profit under uniform pricing	$\pi = 48 - 21 = \\$27\text{m}$

	Profit
Under price discrimination (Q. B1(b)(iii))	\$29m
Under uniform pricing (this legislation)	\$27m
Impact of the legislation	Profit FALLS by \$2m

Why profit falls: this is the standard, general result in price discrimination theory: a firm that is able to price discriminate can always replicate the uniform-pricing outcome if it wanted to (by simply choosing to charge the same price in both markets), so its profit-maximising choice under discrimination can never be lower than under a forced uniform price — discrimination weakly dominates uniform pricing from the firm's perspective. Prohibiting price discrimination by legislation therefore weakly reduces the discriminating firm's profit, exactly as confirmed by the \$2m fall calculated here.

A broader point worth noting for a complete answer: the effect of banning price discrimination on *total welfare* (not just Intel's profit) is theoretically ambiguous in general — it depends on whether total output rises or falls. In this specific case, total output stayed exactly the same (8m units either way), so the main effect of the ban is a pure redistribution: consumers in Japan gain (lower price, more output), consumers in the US lose (higher price, less output), and Intel's shareholders lose (lower combined profit).

Mark Allocation — Part (i) (10%)

Tier	Marks	What separates this tier
8-10%	Correctly constructs the combined demand curve for the price range where both markets are served, correctly solves $MR=MC$ for $Q_{\text{total}}=8$ and $P^*=\$6\text{m}$, and verifies both markets remain served.	
4-7%	Correct general approach but the combined demand curve construction or the verification step is missing.	
0-3%	Incorrect or missing combined demand curve.	

Mark Allocation — Part (ii) (20%)

Tier	Marks	What separates this tier
16-20%	Correctly computes profit under uniform pricing (\$27m), correctly compares it to the price-discrimination profit (\$29m) from part (b), and explains why discrimination weakly dominates uniform pricing in general.	
8-15%	Correct profit calculation but the comparison/explanation of why profit fell is thin.	
0-7%	Incorrect profit calculation.	

Section B · Question 2

Monopoly, Price Discrimination & the Stages of Production · 15+15+40+30%

B2(a)**15%**

Outline the main causes or reasons monopoly arise.

B2(b)**15%**

Outline the conditions necessary for price discrimination to exist.

B2(c)**40%**

Distinguish, using an illustration for each degree, between first-degree, second-degree and third-degree price discrimination. Provide an example of each degree of price discrimination.

B2(d)**30%**

Illustrate the three stages of production and explain the interactions between the Average Product (AP), Marginal Product (MP) and Total Product (TP).

(a) Causes of monopoly — 15%**Main causes of monopoly**

- **Economies of scale (natural monopoly):** if average costs keep falling as output rises across the entire relevant range of demand, a single large firm can supply the market more cheaply than several smaller firms could — entry by a smaller rival would be unable to match the incumbent's costs.
- **Control of a key input or resource:** exclusive ownership or control of an essential raw material, location, or technology that competitors cannot access.
- **Patents and legal barriers:** government-granted exclusive rights (patents, copyrights) or licences/franchises that legally prevent competitors from entering.
- **Network effects:** a product or platform becomes more valuable to each user as more people use it, making it very difficult for a smaller rival network to compete once an incumbent reaches critical mass.
- **High capital requirements / sunk costs:** industries requiring very large upfront investment (e.g. utilities, telecoms infrastructure) deter entry by all but the largest potential competitors.
- **Mergers and acquisitions:** a firm can acquire or merge with its rivals until it controls the entire market, consolidating what was a competitive market into a monopoly.

(b) Conditions for price discrimination — 15%**Conditions necessary for price discrimination to exist**

- **The firm must have some market/monopoly power.** A price-taking firm in a perfectly competitive market cannot price discriminate — it must be a price-maker with some ability to set price above marginal cost.
- **The firm must be able to identify or segment consumers with different price elasticities of demand.** Price discrimination relies on charging different prices to groups (or units) that differ in how sensitive they are to price.
- **Resale (arbitrage) between segments must be preventable.** If a consumer

who buys at the low price could resell to a consumer facing the high price, the price differential would be competed away — effective price discrimination requires segments to be kept separate (e.g. through geography, timing, personal identification, or the nature of the service itself).

Connecting to Q. B1 on this paper: Intel's Japan/US chip pricing (Q. B1(b)) satisfies all three conditions exactly — Intel has market power (it can set prices above marginal cost in each market), can identify and separately price to each national market, and resale between a Japanese and a US computer manufacturer is impractical, keeping the two markets segmented.

(c) The three degrees of price discrimination — 40%

First-degree (perfect) price discrimination

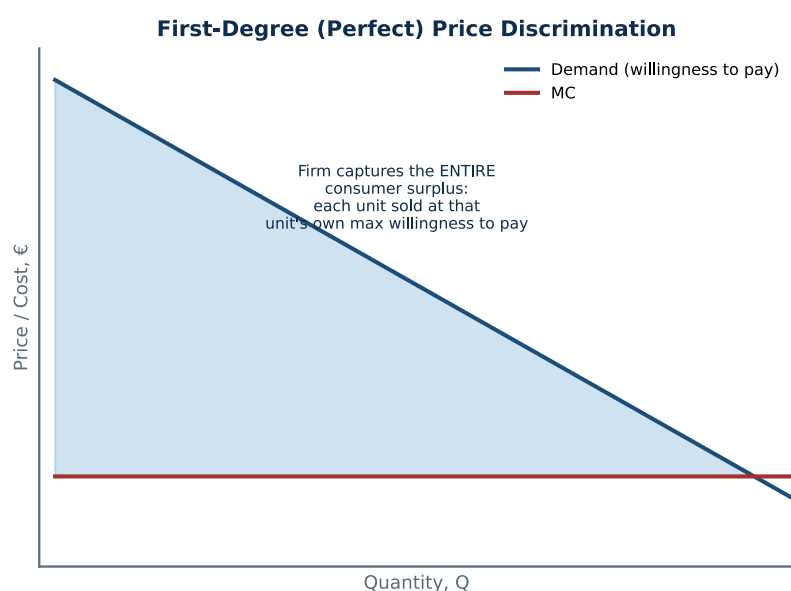


Fig. — First-degree PD: the firm captures the entire consumer surplus

What it is: the firm charges each individual consumer their exact maximum willingness to pay for each unit — extracting the entire consumer surplus as producer surplus, with no deadweight loss (output expands to the competitive/efficient level).

Example: a car salesperson negotiating an individual price with each buyer based on what they read of that buyer's willingness to pay; a personalised/algorithmic online pricing system charging each customer a different price based on their browsing history and data profile.

Second-degree price discrimination

Second-Degree Price Discrimination

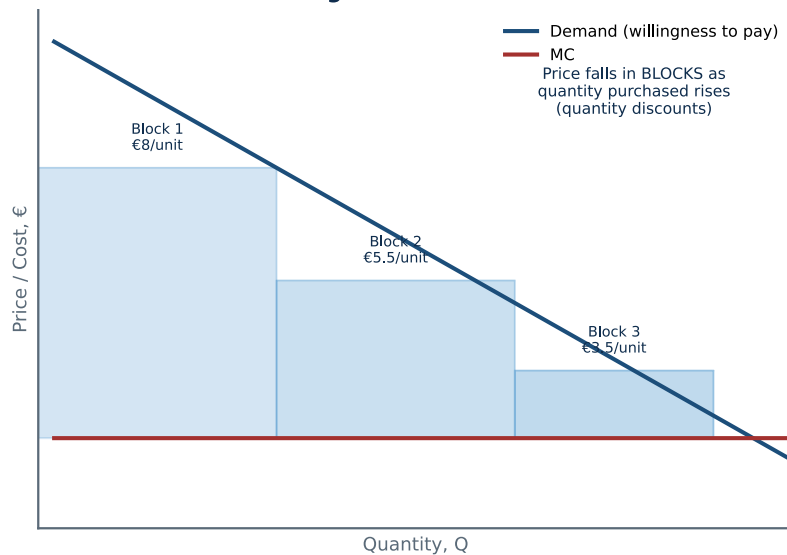


Fig. — Second-degree PD: price varies with the quantity purchased (block pricing)

What it is: the firm charges different prices depending on the quantity a consumer buys (quantity discounts/block pricing), rather than requiring the firm to identify who each individual consumer is — consumers self-select into different blocks based on their own demand.

Example: utility bills charging a lower per-unit rate for electricity/water consumed beyond a certain threshold; bulk-buy discounts (buy 2, get 1 half price); tiered SaaS subscription plans (Basic/Pro/Enterprise) with different quantities/features bundled at different per-unit prices.

Third-degree price discrimination

Third-Degree Price Discrimination

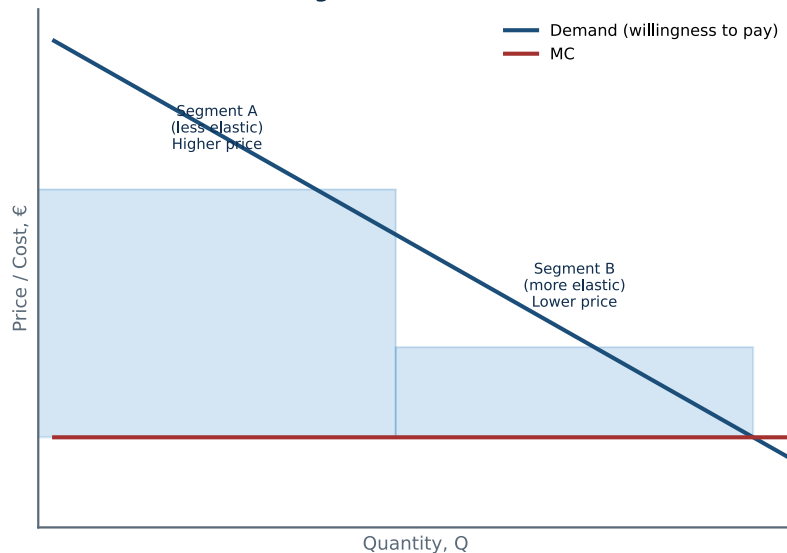


Fig. — Third-degree PD: price varies across identifiable consumer segments by their elasticity of demand

What it is: the firm charges different prices to different, identifiable groups of consumers, based on differences in each group's price elasticity of demand — the group with less elastic (more price-inelastic) demand is charged the higher price, and the group with more elastic demand is charged the lower price (exactly the pattern seen in the Intel Japan/US case, where Japan's less elastic demand was charged the higher price).

Example: student and senior citizen discounts on public transport and cinema tickets; regional/country-specific pricing for the same software or media subscription; off-peak vs. peak pricing for train tickets.

(d) The three stages of production — 30%

The three stages of production

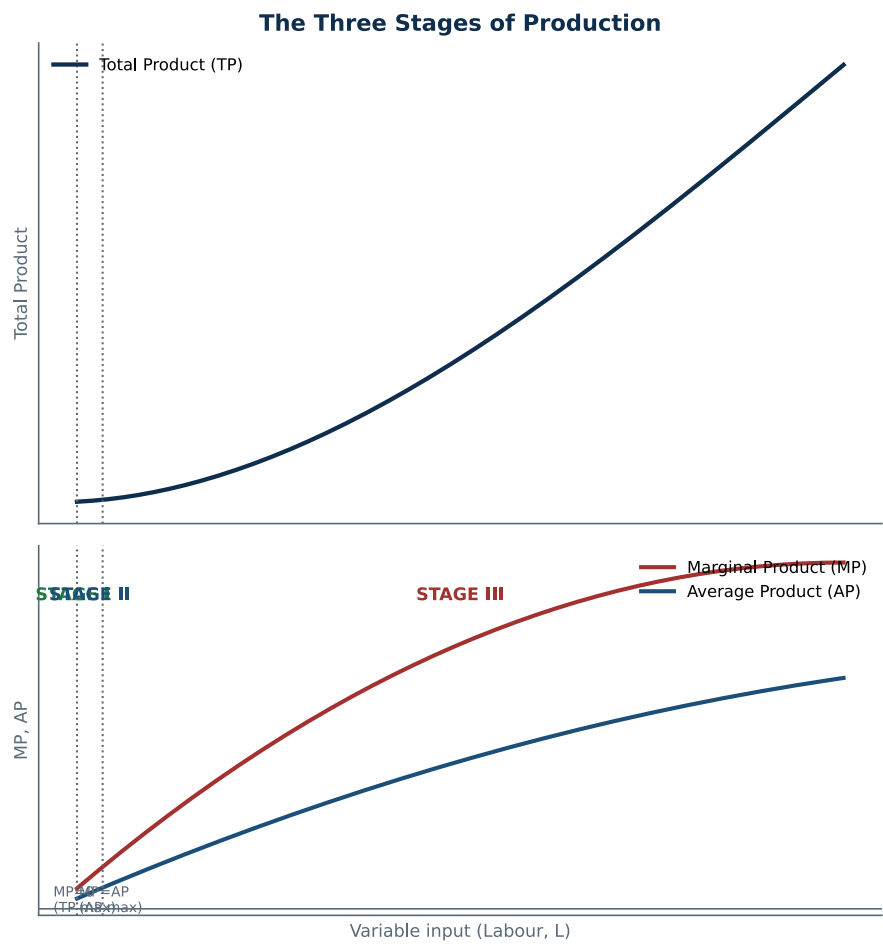


Fig. — Total, Marginal, and Average Product across the three stages of production

Stage	Boundaries	What's happening
Stage I	From zero input up to where $MP = AP$ (AP at its maximum)	MP is rising and exceeds AP, pulling AP up; each additional unit of the variable input adds more to output than the average unit before it — increasing returns.
Stage II	From $MP=AP$ up to where $MP = 0$ (TP at its maximum)	MP is falling (diminishing marginal returns have set in) but is still positive and below AP, so AP is falling too; TP is still rising, just at a decreasing rate. This is the economically rational stage in which to operate — the variable input is still adding to output, and the firm can choose the specific point within this stage using marginal analysis ($MR=MC$).
Stage III	Beyond $MP = 0$	MP is negative — adding more of the variable input actually reduces total output (e.g. overcrowding a fixed amount of capital/land with too much labour). No rational firm would operate here, since less input would produce more output.

The AP/MP relationship (the key interaction the diagram illustrates): whenever MP is above AP, AP is rising (Stage I); whenever MP is below AP, AP is falling (Stages II-III); MP crosses AP exactly at AP's maximum point — the same mathematical relationship as between a marginal and average curve anywhere in economics (e.g. marginal and average cost).

Why Stage II is where a profit-maximising firm operates: in Stage I, the marginal product of the variable input is still rising, meaning the firm isn't yet using its fixed inputs intensively enough — it should keep adding the variable input. In Stage III, marginal product is negative, so the firm is actively using too much of the variable input. Only in Stage II does adding the variable input contribute positively but at a diminishing rate, which is exactly where the marginal-analysis ($MR=MC$) condition for profit maximisation can select a genuine interior optimum.

Mark Allocation — Part (a) (15%)

Tier	Marks	What separates this tier
12-15%	At least four distinct causes of monopoly correctly identified and explained.	
6-11%	Two to three causes correctly identified.	
0-5%	Fewer than two causes, or incorrect.	

Mark Allocation — Part (b) (15%)

Tier	Marks	What separates this tier
12-15%	All three conditions correctly identified and explained.	
6-11%	Two of three conditions correctly identified.	
0-5%	One or none correctly identified.	

Mark Allocation — Part (c) (40%)

Tier	Marks	What separates this tier
30-40%	All three degrees correctly distinguished, each with a correct diagram and a relevant, correctly-classified example.	
18-29%	Two of the three degrees correctly distinguished.	
0-17%	One or none correctly distinguished.	

Mark Allocation — Part (d) (30%)

Tier	Marks	What separates this tier
22-30%	Correct three-stage diagram with the $MP=AP$ and $MP=0$ boundaries correctly identified, and a clear explanation of the AP/MP interaction.	
12-21%	Correct general shapes but the stage boundaries or AP/MP interaction not clearly explained.	
0-11%	Minimal or incorrect treatment.	

Question B2 Total: 100%